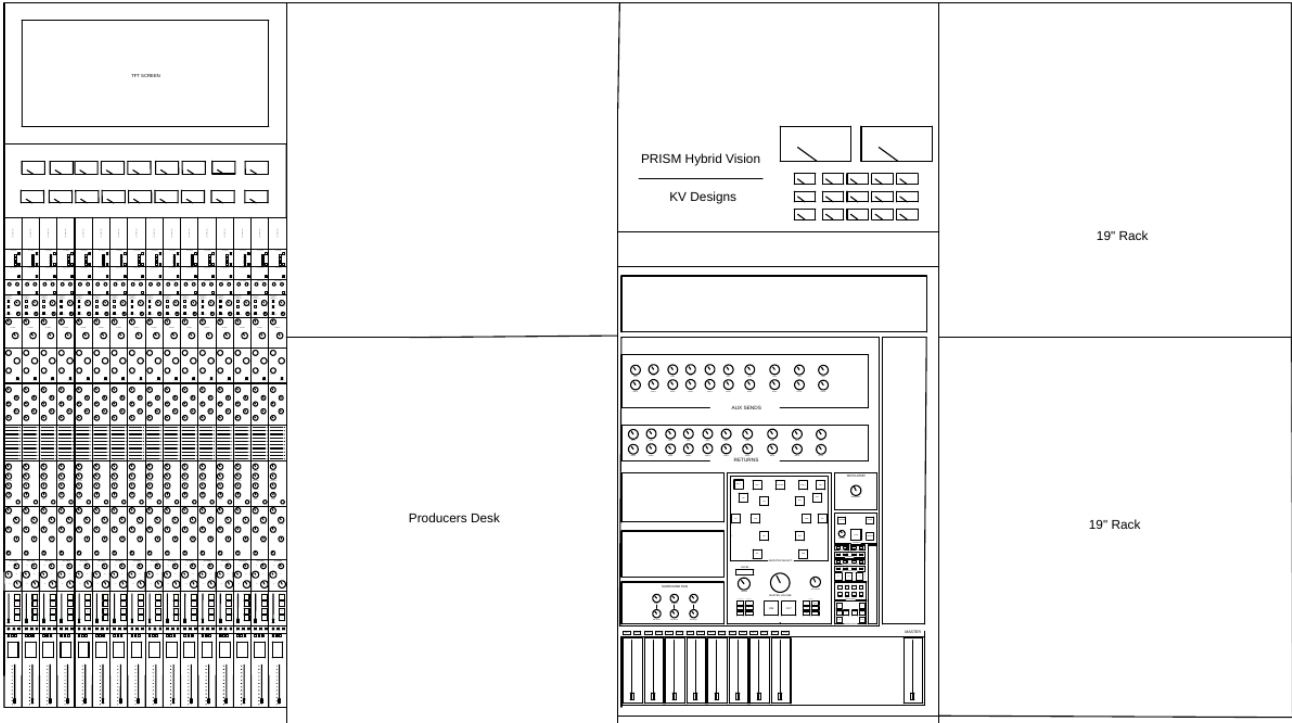


PRISM Hybrid Vision

Technical Document

Prototype 1A



Schematic Walk-through

Channel Strip

Preamp

J82 is the XLR jack that is soldered to the motherboard backplane of the channel strip. This is for modularity. J82 leads to a modular 60 pin connector on the backplane. This connector leads to the modular preamp strip. This strip leads to 51 ohm resistors, and then they lead to T2. This amplifies the voltage. After the transformer, this leads to K10 relay. This could go to the NEVE path or API path. On the API path, it starts with -20dB Pad. The resistors change the dB of the audio signal. This leads to API3 Transformer. After the input transformer, comes a all discrete op amp that will be custom fabricated which is decoupled on the board. This output leads to the output transformer. This leads to K11 relay. This relay switches between the neve path or API path. For the NEVE path, this goes through the neve input transformer, (NEVE2), this amplifies it into the neve op amp. All these paths lead to K12. K12 switches between the tube preamp and skipping the preamp. All this sums up to U175 (PGA2311U). After VOUTL and VOUTR, this leads to the meter and then to K13 relay. K13 switches between U177A and the custom made API discrete op-amp. These opamps convert this to an unbalanced signal. This goes back to the preamp connector and out to the motherboard backplane.

Filters

This goes to another connector to goes to U182A. After U182A, it goes to SW7, the inverse polarity switch. This then goes to the TRIM2 pot. Pin 2 of the pot goes to the non inverting input (+) of the U39B. After U39B, this goes to a feedback loop and to the non inverting input of U183A. After the

output of U183A, goes to RVHIGH2 pot, then goes to the non inverting input of U40B. This is the output of the filter.

Equalizers

The output of the filters goes to the motherboard backplane and to the 5-band equalizer connector.

After the connector, the unbalanced signal goes through BASS, Semi-Bass, Mid, Semi-Treble, and Treble. These are all 10k ohm pots. This then goes through the 10 band equalizer.

Compressors

After the Equalizers, this goes back to a connector on the modular backplane. This backplane goes to an opto compressor connector. After the opto compressor, the output goes immediately to the input of the VCA compressor. After the VCA compressor, it goes to a fader on the channel.

Output

This fader is connected to a touch sensor. This fader then goes to the SUB_Bus button. This switches between group busses. When activated this group bus goes to a copper track across the console. This goes to the sub bus faders on the console. This then goes to a relay K14 which switches between aux sends and the output amplifier, (U202). After U202, this goes to the Output DB25 connector.

This DB25 connector goes to a patch bay or summing mixer; it is for every 8 channels.

Summing Mixer

After the DB25 connector, the op amp converts these signals to unbalanced lines. After the op amp these convert to Stereo channels. These go to Left and right op amps. These all go through a full wave

rectifier into a ADC converter (U52). After U52, DATA/DSDR, this leads to SDATA_IN0 of the Sigma DSP processor, this then goes to a DAC converter. This then goes to the AUX send relays; which switch between AUX sends and to a stereo output. The stereo output leads to the ADC converter. The output of the DAC leads to the LEFT and RIGHT inputs of the monitoring section of the summing mixer (J8 and J7). This leads to the ADC, this then goes to the SigmaDSP processor, after the processor this leads to 74HC595's, which then leads to DACs that go to PGA23s which each go to faders which lead to the xlr output on the backside of the console.

Revisions

RV1: Changed fixed to modular interface (uAMX).

RV2: Changed aux sends in summing mixer to before monitoring section.

RV3: Chaned AUX sends from TRS to XLR.

Current Unknowns

- Daisy chaining PGA2311U may be wrong/incorrect
- Electrical Noise in the circuitry

Failure Modes

- Capacitors are rated under 400V
- Thermal issues
- dense circuitry